

CariSurg MedTech Pathways Research Project

Project Title: Evaluation of a Non-Autonomous AI Decision-Support Tool for Standardizing Emergency Triage Intake and Managing Missing Biometric Data Dynamics at Mercer General Hospital

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Statement of the Problem:

Drawing on my experience as a Caribbean Medical Assistant, this project focuses on findings reported by Eriten (2024) regarding the potential of artificial intelligence (AI) to support emergency department triage. Human variability in recording vital signs may contribute to delays and inconsistencies during periods of high patient volume at Mercer General Hospital. Eriten (2024) also highlights concerns regarding reduced model performance when biometric data are incomplete. To address this operational challenge within regional resource constraints, this study proposes a 12-week pilot evaluation of a non-autonomous AI decision-support tool aimed at standardising triage intake processes and assessing routing performance under simulated high-demand conditions.

Previous Work:

Substantial academic work within the last decade has explored the use of artificial intelligence and machine learning to reduce emergency department overcrowding, improve triage consistency, and address technical barriers associated with healthcare data systems.

Literature Summaries:

1. Eriten, S. (2024). Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives

Problem: The study examines how AI-supported triage systems may help address emergency department overcrowding and variability in clinical prioritisation.

Method: The author reviews contemporary machine-learning approaches applied to emergency triage and evaluates their reported performance across multiple healthcare settings.

Outcome: The review suggests that AI models can improve consistency in patient prioritisation and demonstrate promising predictive performance when trained on high-quality datasets.

Limitation: The study notes that model performance may decline when key clinical variables or biometric measurements are missing from patient records.

2. Picard, J. (2023). Measuring inter-rater variation and triage-level disagreement across high-volume acute care services

Problem: The study investigates variation in triage decisions among healthcare professionals working in acute care environments.

Method: A retrospective analysis was conducted to evaluate agreement rates among clinicians assigning triage categories.

Outcome: Findings indicate that differences in clinical judgement can contribute to inconsistencies in patient prioritisation and may affect operational efficiency.

Limitation: The study was conducted within NHS settings and therefore may have limited generalisability to resource-constrained Caribbean healthcare environments.

3. Tyler, R., Chang, H., & Mensah, J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Global Scoping Review of Algorithmic Architectures and Validation Gaps

Problem: The review examines the development, implementation, and validation of AI-assisted triage systems in hospital emergency departments.

Method: The authors conducted a global scoping review of published studies investigating machine-learning approaches to triage.

Outcome: The review found growing evidence that AI may support emergency department workflow efficiency and patient prioritisation.

Limitation: Most validation studies were conducted in high-resource settings, highlighting a lack of evidence from low- and middle-income regions.

4. Araouchi, Y., & Adda, M. (2024). Multimodal Fusion Architectures for Incomplete Clinical Charting Data: Evaluating the TriageIntelli Framework

Problem: The study explores methods for maintaining triage model performance when clinical records contain incomplete information.

Method: The authors developed and evaluated a multimodal fusion architecture designed to process incomplete patient datasets.

Outcome: Results indicated that the framework maintained relatively stable classification performance when selected non-critical variables were unavailable.

Limitation: Evaluation was conducted using simulated datasets, limiting conclusions regarding performance in real-world clinical environments.

5. Da'Costa, N., Emmanuel, J., & Reynolds, S. (2025). Deep Learning Triage Validation and Legacy EHR Interoperability Bottlenecks Across Regional Healthcare Networks

Problem: The study investigates challenges associated with implementing AI-supported triage systems in healthcare environments with fragmented electronic health record infrastructure.

Method: Deep-learning triage models were evaluated within regional healthcare networks to assess data exchange and interoperability challenges.

Outcome: The findings identified several technical barriers that may affect the reliability and efficiency of AI-supported triage workflows.

Limitation: The study focused primarily on technological infrastructure and did not examine frontline clinician adoption or workflow integration.

6. De Freitas, L., Goodacre, S., O'Hara, R., Thokala, P., & Hariharan, S. (2020)

Problem: The study investigated factors affecting patient flow within a Caribbean emergency department and identified operational barriers contributing to delays in patient care.

Method: Researchers conducted a qualitative study using interviews and thematic analysis to explore staff perspectives regarding emergency department workflow and patient movement.

Outcome: The study identified resource limitations, communication barriers, and workflow inefficiencies that negatively affected patient throughput and emergency department operations.

Limitation: The study focused on a single Caribbean emergency department and did not evaluate artificial intelligence or digital decision-support systems as potential solutions.

7. Moor, M., et al. (2023)

Problem: The study examined the potential of foundation models to support a wide range of clinical decision-making tasks across healthcare environments.

Method: The authors reviewed emerging foundation-model architectures and evaluated their applications in prediction, diagnosis, and clinical decision support.

Outcome: Results suggested that foundation models may improve adaptability and performance across multiple healthcare tasks while reducing the need for task-specific model development.

Limitation: The study highlighted concerns regarding data quality, transparency, bias, and implementation challenges in real-world healthcare settings.

Literature Gap Analysis:

Research Gap 1: Model Performance Under Incomplete Clinical Data Conditions

Evidence and Justification

- Existing literature demonstrates that missing clinical information can negatively affect the performance of AI-assisted triage systems.
- Eriten (2024) and Da'Costa et al. (2025) identify data quality and completeness as important determinants of model reliability.
- Limited evidence exists regarding the performance of triage decision-support tools in resource-constrained healthcare settings where incomplete patient records may occur more frequently.
- Additional validation is therefore needed within Caribbean clinical contexts.

Research Gap 2: Limited Evidence from Caribbean and Other Resource-Constrained Settings

Evidence and Justification

- Tyler et al. (2024) report that most AI-triage validation studies have been conducted in high-resource healthcare systems.
- Picard (2023) focuses on NHS-based environments with established technological infrastructure and staffing models.
- There is limited evidence regarding the effectiveness, implementation requirements, and operational challenges of AI-supported triage tools in Caribbean emergency departments.
- De Freitas et al. (2020) demonstrated operational and workflow challenges within a Caribbean emergency department but did not investigate whether AI-assisted triage systems could help address these constraints. This represents an important opportunity for future research within regional healthcare environments.

Proposed Solution and Alignment:

To address these identified gaps, this project proposes a 12-week pilot study evaluating a non-autonomous, human-in-the-loop AI decision-support tool tailored to the operational environment of Mercer General Hospital.

Direct Literature Alignment:

1. Human-in-the-Loop Safeguards

- The system will provide recommendations only.
- Healthcare professionals will retain full responsibility for clinical decision-making.
- This approach responds directly to concerns regarding safety and governance identified by Tyler et al. (2024).

2. Missing Data Resilience Testing

- The pilot incorporates multimodal fusion principles informed by Araouchi and Adda (2024).
- Simulated patient records with missing biometric data will be used to assess algorithmic stability.

3. Zero-Risk Simulation Environment

- The pilot will use synthetic datasets rather than live hospital records.
- This approach addresses interoperability concerns identified by Da'Costa et al. (2025).
- It also keeps the project achievable within the 12-week timeframe.

Project Objectives:

1. **Reduce inter-rater triage scoring variability** by at least 15% across simulated patient intake scenarios.
2. **Maintain a classification accuracy of at least 90%** when up to two non-critical biometric variables are absent from patient records.
3. **Develop a comprehensive operational implementation playbook** outlining workflow procedures, data-management requirements, risk-mitigation strategies, and recommendations for future deployment within Caribbean healthcare settings.

References :

1. Eriten S. Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives. *Turk J Emerg Med.* 2024;24(1):12-25.
2. Picard J. Measuring inter-rater variation and triage-level disagreement across high-volume acute care services. *Ann Emerg Med.* 2023;82(4):411-423.
3. Tyler R, Chang H, Mensah J. Use of artificial intelligence in triage in hospital emergency departments: A global scoping review of algorithmic architectures and validation gaps. *Acad Emerg Med.* 2024;31(3):245-258.
4. Araouchi Y, Adda M. Multimodal fusion architectures for incomplete clinical charting data: Evaluating the TriageIntelli framework. *Int J Med Inform.* 2024;172:105-118.

5. Da'Costa N, Emmanuel J, Reynolds S. Deep learning triage validation and legacy EHR interoperability bottlenecks across regional healthcare networks. *Caribb J Digit Health*. 2025;3(1):88-102.
6. De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open*. 2020;10(12):e041422. doi:10.1136/bmjopen-2020-041422.
7. Moor M, Banerjee O, Abad ZSH, et al. Foundation models for generalist medical artificial intelligence. *Nature*. 2023;616(7956):259-265. doi:10.1038/s41586-023-05881-4.